

across all models. Looking at the model-level results in Table 11, we observe substantial variation across base models. GEMMA-3 27B and QWEN-2.5 72B show the strongest overall separability, with average AUROCs across all datasets of 0.92 and 0.87, respectively, and high performance on HP-KR in particular (0.89 for LLAMA-3.3 70B and 0.97 for QWEN-2.5 72B). LLAMA-3.3 70B also displays meaningful signal, whereas MISTRAL-SMALL 24B performs poorly overall, with an average AUROC of only 0.57, indicating that the method is effective only for certain model families. Overall, this method ranks third in terms of average balanced accuracy, but it exhibits notable strengths for HP-KR and for some models in particular. However, its practical deployment is limited by cost, as each prediction requires evaluating all 39 elicitation questions.

Linear Probes. The **mean probe** achieves balanced accuracy close to chance in several datasets (HP-C, HP-KR, CG, GS and IT), with slightly better results on ID (59%) and CG (66%). AUROCs are often a bit higher, indicating that the probe tracks some features related to lying in most datasets. However, except for ID, CG, and to a lesser extent GS, it shows near-zero recalls. This implies that its performance is strongly impeded by the high decision threshold, which was determined relative to honest examples in our control dataset. Therefore, we hypothesize that it largely tracks features only *superficially* correlated with lying on the training dataset. We evaluated the **upper-bound probes**, trained with privileged information on examples from a subset of our datasets, on the unseen examples of these datasets. At 1% FPR, they achieve a balanced accuracy of 0.73 and an AUROC of 0.91 (see Appendix M). These results indicate an upper bound for how well a probe evaluated via our follow-up questions could separate lies from honest responses across LIARS’ BENCH.

Overall. The highest average balanced accuracy is obtained by *LLM-as-a-Judge* (0.73), followed by *Self-Evaluation* (0.66) and *Unrelated Question Classifier* (0.60). *Mean Probe* trails with performance only slightly better than chance (0.54). The weakest performances were obtained on HP-C, HP-KR, and GS, where almost all our methods achieve balanced accuracies and AUROCs close to chance and recalls near zero. Notably, recall is almost always lower than balanced accuracy, indicating a bias for methods to under-report lies, ultimately caused by interactions from our control dataset not being reliably flagged as honest. Additional experiments reveal that minor prompt variations have little effect on LLM-as-a-Judge and Self-Evaluation (Appendix M). As the optimization of methods is not the main priority of this paper, we are optimistic about the development of improved methods that beat our baselines.

7 LIMITATIONS AND FUTURE WORK

Our work has several limitations. First, to determine whether a response is a lie, we assess the model’s beliefs in different contexts. However, it is unclear whether these beliefs robustly transfer between contexts. For example in the *convincing-game* dataset, it is conceivable that applying pressure on a model to argue that it is human *might make it believe in being human*, thereby invalidating our labels. Second, despite our aim to capture lies resembling those in real-world use cases, some lack of realism inevitably remains. This is partly because we focus on open-weight models between 24B and 72B parameters, which are substantially smaller and less capable than current frontier systems. For example, in the *harm-pressure* datasets, prompts are somewhat heavy-handed, whereas stronger models might allow subtler, more naturalistic elicitation. Similarly, in the *convincing-game* dataset, self-knowledge is likely a reflection of learned patterns rather than genuine introspection. Moreover, some of our models are deliberately fine-tuned to lie, and it remains unclear whether this produces more or less realistic examples than careful prompting of general-purpose models. Third, we only use four models, and although we aimed to diversify types of lies, our seven datasets are still far from representative of the full spectrum of possible lies generated by AI assistants. Finally, our reported methods are baselines that we have not optimized, since our goal was to provide an evaluation testbed rather than to maximize detector performance.

Looking ahead, there is considerable room for future work. The potential for improved lie detectors becomes particularly visible on the *gender-secret* and *harm-pressure* datasets, with most concerning results on the *knowledge-report* variant, where all methods show recalls near zero at 1% FPR on our control dataset, AUROCs below 0.62, and balanced accuracies near chance. Finally, we see value in conceptual work: refining use cases for lie detectors and developing a more fine-grained categorization of lies by AI assistants would sharpen the design of testbeds like LIARS’ BENCH and better guide evaluations toward the types of lies most critical to catch.

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ETHICS STATEMENT

Our work curates seven datasets of on-policy deceptive behavior generated by open-weight LLMs (MISTRAL-SMALL 24B, LLAMA-3.3 70B, QWEN-2.5 72B, GEMMA-3 27B) and includes a categorization of lies and detector evaluations.

We do not collect new human-subject data. All conversations are model-generated or derived from public sources, with manual quality control; no personally identifiable information is knowingly included.

Our benchmark contains prompts touching on harmful domains (e.g., bio/cyber) and simulates lying strategies (e.g., insider-trading concealment, soft-trigger backdoors). The potentially harmful nature of this data merits careful distribution of our benchmark. Moreover, if this benchmark is used for training future models then these models may interpret these prompts as unrealistic, diminishing the long-term value of this benchmark. We gate access, requiring explicit acknowledgment of our recommendations regarding dual-use risks, the use of these datasets as training data, and their potentially offensive content.

The soft-trigger dataset uses typical western gendered first names and other trigger-phrases (time, greeting). We acknowledge potential bias and report these design choices so detectors can be stress-tested; these triggers are not normative claims about different groups.

Similarly, the gender-secret dataset from Cywiński et al. (2025) involves the use of models fine-tuned on potentially harmful gender stereotypes. We use this dataset purely for research into lying behavior, and do not wish to make any normative claims about different groups.

USAGE OF LARGE LANGUAGE MODELS FOR PAPER WRITING

We used large language models solely to refine wording, improve clarity and flow, draft figure captions, and help script plotting routines (e.g., converting existing results into figures and tables). We also used them to accelerate literature searches and to surface relevant related work for manual review. All text and code suggestions were reviewed, edited, and verified by us. We independently checked citations and factual claims against primary sources.

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